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JAVA PROJECT REPORT ON

Savings Goal Planning and Financial Progress Tracking System

Submitted by

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Under the Guidance of

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Semester: **I**

DECLARATION

programming is a **record of my original work** carried out under the guidance of **Dhileeban A**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

_____

JANANI .N

CERTIFICATE

[**Department Name]**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **[java programming]**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: Coimbatore

Date:09.04.2026

Submitted for the Viva-Voce Examination held on _____

Guide:
(Technical Trainer, KGCAS)

Head of the Department
(Technical Head, KGCAS)

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ABSTRACT

The Savings Goal Planning and Financial Progress Tracking System is a console-based Java application designed to automate product inventory handling and billing operations in small retail shops. In traditional retail environments, inventory management and billing processes are usually performed manually using notebooks or calculators. These manual operations often result in incorrect calculations, inefficient stock tracking, and time delays during customer billing.

This project provides a computerized solution to manage product details such as product ID, product name, price, and quantity available in stock. The system allows the shopkeeper to add products into the inventory, view available product details, and generate customer bills by selecting products and specifying purchase quantities. The system automatically calculates the total bill amount and updates product stock after each transaction.

The application is implemented using **Core Java programming concepts**, including classes, objects, ArrayList collections, loops, conditional statements, methods, and user input handling using the Scanner class. The project demonstrates the practical application of object-oriented programming concepts in solving real-world business problems.

The primary objective of this system is to reduce manual effort, improve billing accuracy, manage stock efficiently, and provide faster service to customers. This project can be further enhanced by integrating database connectivity and graphical user interface support.

1. INTRODUCTION

Inventory management and billing are two essential operations in every retail shop. Maintaining accurate records of product availability and generating bills efficiently are important for smooth business operations. In small-scale retail shops, these operations are often performed manually, which can lead to errors in calculations and difficulties in maintaining stock records.

With the advancement of computer technology, automated inventory and billing systems have become necessary for improving operational efficiency and minimizing human errors. A computerized inventory and billing system helps shopkeepers maintain accurate product records, track stock availability, and generate bills quickly and accurately.

The Savings Goal Planning and Financial Progress Tracking System is

developed as a Java-based console application that simplifies inventory management and billing processes. The system provides options for adding products, viewing inventory details, and generating bills for customers. It automatically calculates the total price of purchased items and updates product quantity after billing.

This project demonstrates how Core Java programming concepts can be applied to develop a simple real-world application that supports retail shop operations efficiently

1.1 Objective

The main objectives of this project include:

- To design and develop a retail shop inventory system using Java
- To maintain product information such as ID, name, price, and quantity
- To provide a menu-driven interface for easy system interaction

• To automate the billing process in retail shops
• To calculate total bill amount automatically
• To reduce errors caused by manual billing methods
• To update inventory automatically after product purchase
• To demonstrate the implementation of object-oriented programming concepts
• To improve operational efficiency in small retail environments

1.2 Overview

The Savings Goal Planning and Financial Progress Tracking System is a menu-

driven console application developed using Java programming language. The system maintains a collection of products stored in memory using the ArrayList data structure.

The application provides three main functionalities:

1. Adding products into inventory
2. Viewing stored product details
3. Generating customer bills

The shopkeeper can add product details such as product ID, product name, product price, and quantity available in stock. The system stores these details in memory and allows users to retrieve them whenever required.

During billing, the user enters the product ID and quantity required by the customer. The system calculates the total price automatically and updates the remaining stock quantity accordingly.

The application is simple, user-friendly, and suitable for small-scale retail shop management.

1.3 Features

The system includes the following important features:

Product Management

Allows adding new products into the inventory with necessary details.

Inventory Viewing

Displays complete list of available products along with price and quantity.

Billing Automation

Automatically calculates the total bill amount based on selected products.

Stock Updating

Reduces product quantity automatically after each purchase.

Error Handling

Displays appropriate messages if product ID is invalid or stock is insufficient.

Menu-Driven Interface

Provides simple numbered options for easy user interaction.

2. PROGRAM ANALYSIS

Program analysis is the process of understanding the requirements of the system and identifying problems in the existing manual system.

In many retail shops, billing and inventory operations are performed manually using registers and calculators. This leads to inefficiency and incorrect stock management.

To overcome these problems, a computerized inventory and billing system is required.

The proposed system automates these tasks using Java programming language.

2.1 Problem Definition

Manual inventory and billing systems create several operational challenges such as:

- Difficulty in tracking available stock
- Incorrect bill calculations
- Time-consuming billing process
- Possibility of human errors
- Lack of organized product records

These problems affect both shopkeepers and customers.

Therefore, a computerized system is required to manage inventory efficiently and generate accurate bills automatically.

The Savings Goal Planning and Financial Progress Tracking System solves these problems by providing a simple automated solution.

2.2 Project Overview

The project is designed as a console-based Java application that maintains product information using an ArrayList collection.

The system supports the following operations:

• Adding product details
• Displaying inventory information
• Selecting products for billing
• Calculating bill amount
• Updating stock after purchase

The program uses object-oriented programming concepts such as:

Class → Product
Object → Individual product records
ArrayList → Storage for inventory
Methods → Functional operations

2.3 Scopes and Limitations

Scope

The system can be used in:

• Grocery shops
• Medical stores
• Stationery shops
• Small supermarkets
• Departmental stores

Advantages include:

• Faster billing process
• Accurate stock management
• Easy product handling
• Reduced manual effort

Limitations

Despite its advantages, the system has some limitations:

• Data is stored temporarily (no database support)
• System supports only single user
• No graphical user interface available
• Cannot generate printed invoice

Future improvements can overcome these limitations

3. SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before the actual implementation. In this phase, the system is divided into different modules, and the workflow of the application is defined. Proper system design helps in organizing the program efficiently and makes development easier.

The Savings Goal Planning and Financial Progress Tracking System is designed as a menu-driven application where users can select food items, enter quantities, and generate a final bill automatically.

The system is divided into several modules to perform specific tasks. Each module is responsible for handling a particular function of the system.

3.1 Modules

The main modules of the Savings Goal Planning and Financial Progress Tracking System are explained below:

1. Menu Display Module

This module is responsible for displaying the list of available food items along with their prices. The menu helps the user select the required items for billing.

Example Menu:

1. Veg Fried Rice ₹120
2. Chicken Biryani ₹200
3. Paneer Butter Masala ₹180
4. Butter Naan ₹40
5. Cold Drink ₹30

The menu is displayed when the program starts so that the user can easily view the available options.

2. Order Selection Module

This module allows the user to select food items from the menu and enter the quantity ordered by the customer.

Functions of this module include:

• Accepting user input for item selection
• Accepting quantity of selected items
• Storing order details temporarily

The system allows users to select multiple food items before generating the final bill.

3. Bill Calculation Module

This module performs all the billing calculations.

The system calculates:

• Total price of each item
• Subtotal of all selected items
• Applicable tax (GST)
• Final payable amount

The calculation is performed automatically using Java programming logic.

Example Calculation:

Item Total = Price × Quantity
GST = 5% of Subtotal
Final Bill = Subtotal + GST

4. Invoice Generation Module

This module generates the final bill for the customer.

The bill includes:

• Item name
• Quantity
• Price
• Subtotal
• GST amount
• Final bill amount

Example Bill Output:

------ HOTEL BILL ------

Item	Qty	Price
Chicken Biryani	2	400
Cold Drink	2	60

Subtotal	460
GST (5%)	23
Total	483

Thank You Visit Again!

5. Exit Module

This module allows the user to exit the application safely after completing the billing process.

When the user selects the exit option, the system terminates the program.

4. METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages:

• Problem Analysis
• System Design
• Program Development
• Testing and Validation
• Result and Documentation

4.1. Problem Analysis

In this phase, the problem is studied and analyzed in detail. The requirements of the system are identified based on the needs of restaurant operations.

The main problems identified in manual billing systems are:

• Manual calculations may lead to billing errors.
• Preparing bills manually takes more time.
• Difficult to handle multiple customer orders efficiently.
• Lack of automated records for transactions.

Based on these problems, the system requirements were defined to develop an automated billing application that simplifies restaurant billing operations.

4.2. System Design

System design defines how the application structure is organized into different modules.

The project follows a modular approach where each module performs a specific task independently.

Product ID
Product Name
Product Price
Product Quantity

The program flow and structure are designed so that users can easily interact with the system.

4.3. Program Development

The system was implemented using Java programming language.

Technologies used:

Core Java
Scanner class
ArrayList class
Loops
Conditional statements
Methods

Two classes were created:

Product class
RetailShopSystem class

4.4. Testing and Validation

Testing was performed with multiple input conditions.

Example scenarios tested:

Valid product entry
Invalid product ID
Insufficient stock condition
Multiple product billing

Results confirmed correct functioning of system.

4.5. Result and Documentation

After successful testing, the results of the system are documented. The system generates the final bill based on the selected food items and quantities.

The results are demonstrated through program execution and screenshots showing the input and output of the system.

Proper documentation is prepared to explain the design, implementation, and working of the system.

4.6. Algorithm

Step 1: Start program
Step 2: Display menu options
Step 3: Accept user choice
Step 4: If choice = Add Product → store product details
Step 5: If choice = View Inventory → display product list
Step 6: If choice = Generate Bill → enter product ID
Step 7: Enter quantity
Step 8: Calculate total price
Step 9: Update stock quantity
Step 10: Display final bill

Step 11: Repeat until exit selected
Step 12: Stop program

4.7. Flowchart

START

↓

Display Menu

↓

Enter Choice

↓

Add Product / View Inventory / Generate Bill / Exit

↓

If Add Product → Enter Details → Store Product

↓

If View Inventory → Display Products

↓

If Generate Bill → Enter ID → Enter Quantity

↓

Calculate Total Bill

↓

Update Stock

↓

Display Bill

↓

Repeat Until Exit Selected

↓

END

5. SOURCE CODE

Include important parts of the Java code such as:

Main class

Menu logic

Product management

Billing calculation

```
// Total calculation
double total = price * quantity;

// GST calculation (5%)
double gst = total * 0.05;

// Final bill
double finalBill = total + gst;
```

6. INPUT AND OUTPUT SCREENS

Include screenshots of:

Program menu (Retail Shop Menu)

Product addition / Item selection

Billing output

Final bill / Invoice

Example:

Figure 1: Retail Shop Menu Screen

Figure 2: Product Entry Screen

Figure 3: Billing Output

Figure 4: Final Invoice

7. DISCUSSION & CONCLUSION

The developed system simplifies retail shop inventory tracking and billing operations. The application successfully demonstrates the practical implementation of object-oriented programming concepts such as classes, objects, and ArrayList collections.

The system provides accurate billing results and reduces manual calculation errors. It also helps shopkeepers monitor available stock efficiently.

Testing confirmed that the system handles both valid and invalid inputs correctly.

7.2 Conclusion

The Savings Goal Planning and Financial Progress Tracking System was

successfully developed using Java programming language.

The system improves inventory tracking efficiency and automates customer billing operations. It reduces manual effort and improves billing accuracy in retail shops.

This project demonstrates how programming concepts learned in Core Java can be applied to develop real-world business applications.

Future enhancements may include:

Database integration
Graphical user interface
Barcode scanning support
Automatic report generation
Multi-user access support

8. REFERENCES

The following resources were used during the development of the Savings Goal Planning and Financial Progress Tracking System project:

Books

• *Herbert Schildt – Java: The Complete Reference*
• *E. Balagurusamy – Programming with Java: A Primer*
• *Y. Daniel Liang – Introduction to Java Programming*

These books helped in understanding Core Java concepts such as classes, objects, loops, methods, and collections.

Websites

• https://www.oracle.com/java
(Official Java documentation for syntax and programming concepts)

• https://www.geeksforgeeks.org
(Used for understanding ArrayList, loops, and Java examples)

• https://www.javatpoint.com
(Reference for Java methods, classes, and Scanner usage)

• https://www.w3schools.com/java
(Basic Java syntax and programming structure)